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Signature of collective elastic glass physics in surface-induced long-range tails in dynamical gradients

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Understanding the underlying nature of dynamical correlations believed to drive the bulk glass transition is a long-standing problem. Here we show that the form of spatial gradients of the glass transition temperature and structural relaxation time near an interface indeed provide signatures of the nature of relaxation in bulk glass-forming liquids. We report the results of long-time, large-system molecular dynamics simulations of thick glass-forming polymer films with one vapour interface, supported on a dynamically neutral substrate. We find that gradients in the glass transition temperature and logarithm of the structural relaxation time nucleated at a vapour interface exhibit two distinct regimes: a medium-ranged, large-amplitude exponential gradient, followed by a long-range slowly decaying tail that can be described by an inverse power law. This behaviour disagrees with multiple proposed theories of glassy dynamics but is predicted by the 'elastically collective nonlinear Langevin equation' theory as a consequence of two coupled mechanisms: a medium-ranged interface-nucleated gradient of surface-modified local caging constraints, and an interfacial truncation of a long-ranged collective elastic field. These findings support a coupled spatially local–nonlocal mechanism of activated glassy relaxation and kinetic vitrification in both the isotropic bulk and in broken-symmetry films.

The underlying mechanism driving the experimentally observed kinetic glass transition has been one of the most enduring and debated open questions in condensed matter physics[1,2]. The challenge is to explain the rapid growth, upon cooling, of the activation energy for structural α-relaxation in 'fragile'[3] glass-forming liquids. Numerous theories have been proposed for this phenomenon[1]. Most differ fundamentally in terms of the structural, thermodynamic or dynamic property and physical mechanism postulated to drive the growth of the apparent activation energy. Proposed origins include loss of configurational entropy[4–6], loss of available free volume[7], percolation of slow-relaxing domains[8], dynamical facilitation among a small subset of mobile particles[9], emergence of glassy elasticity on cooling[10], onset of particle localization[11], and a coupled combination of long-range elasticity and cage-scale localization[12,13], both causally controlled by structural correlations. The core challenge of the field has been an inability to definitively discriminate between these various theoretical approaches.

For 35 years, considerable research has sought to employ confinement and broken symmetry in thin films and near interfaces to reveal the physics of dynamical cooperativity and thus to potentially resolve the nature of the bulk glass transition[14–17]. Moreover, activated spatially

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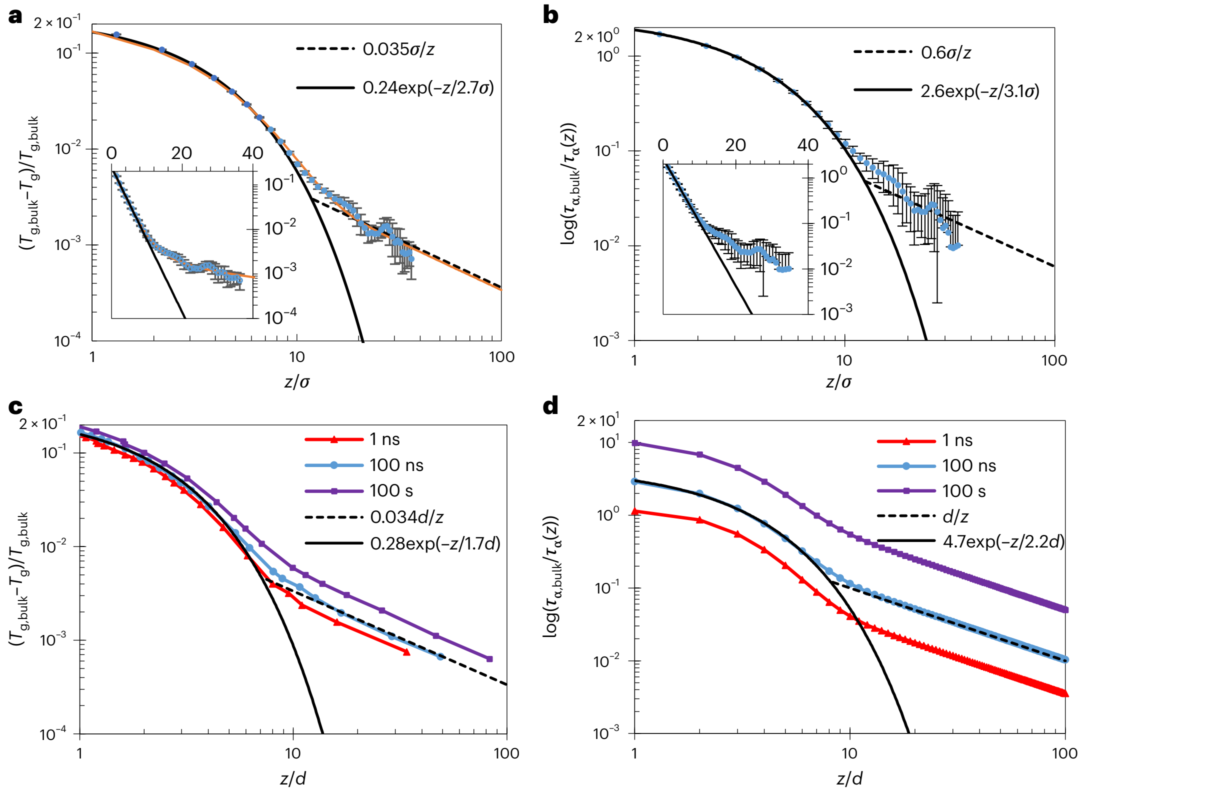

Fig. 1 | Simulation and theory of normalized dynamical gradients of the shift of the glass transition temperature and alpha relaxation time relative to the bulk. a,**b**, Simulated normalized T_g shift gradients (defined at a computational timescale of 10^5 LJ time units) (**a**) and $\log(\tau_{\alpha,\mathrm{bulk}}/\tau_\alpha(z))$ gradients (obtained at a temperature for which the bulk relaxation time is 10^5 LJ time units, using the form of ref. [40] to smooth data across multiple temperatures) (**b**) versus normalized distance z/σ from the free surface. Data are presented as mean values, with error bars reflecting standard deviation values propagated from data for three films and three bulk runs. Solid black lines are fits of low-z data to a decaying exponential. Dashed black lines are an inverse power law, z^{-1}. The solid orange curve in **a** is a fit to equation (6). Insets show the same data for a linear ordinate axis. **c**,**d**, ECNLE theory predictions for a spherical particle model (diameter d) of T_g gradients (**c**) and $\log(\tau_{\alpha,\mathrm{bulk}}/\tau_\alpha(z))$ gradients (**d**) plotted on logarithmic scales versus normalized distance z/d from the interface. Distinct datasets are results at very different choices of the vitrification timescale, as shown in the legends, and lines are exponential and power-law fits to theory results in the low- and high-z limits, respectively, as shown in the legends.

heterogeneous dynamics near interfaces and under confinement is of great fundamental interest in its own right (for example, for surface diffusion[18] and ultra-stable glass formation[19]), and it bears on the behaviour of the wide range of materials that have internal or external dimensions on the nanoscale[20], including films[21], nanocomposites[22], block copolymers[23], semicrystalline materials[24], liquids in pores[25] and many more. It is now established that confinement and proximity to an interface can strongly modify the dynamics of glass-forming liquids, over a considerably longer distance than interfacial or confinement-induced perturbations to thermodynamics and structure[16,17,21,26–28]. A challenge thus far has been a lack of sufficient knowledge of the spatially resolved form of these alterations to definitively distinguish between proposed mechanisms of bulk glass formation and their alteration near interfaces. As reviewed recently[26], specific and distinct predictions of the spatial form of such gradients have been made by competing theories of glassy activated dynamics, including the random first order transition (RFOT) entropy crisis theory[29], free volume percolation theories[8,30], the cooperative free volume model[31], dynamical 'string' theories based on the Adams–Gibbs model[32], and the elastically cooperative nonlinear Langevin equation (ECNLE) theory[33]. The form of interfacial gradients in T_g and relaxation time thus appears to offer an opportunity to genuinely distinguish between competing theories of the glass transition.

In this Article we describe long-time, large-system molecular dynamics simulations of a bead-spring polymer thin film supported on a dynamically neutral substrate, to elucidate the spatial form of the glass transition temperature (T_g) and structural relaxation time gradients out to large distances from a free vapour surface. Our exploration of the long-range nature of this gradient is motivated by recent predictions of ECNLE theory, which is a microscopic, force-based, particle-level theory for activated dynamics in (originally bulk) glass-forming liquids[12,13]. In both the bulk and near interfaces, ECNLE theory predicts two additive and physically distinct, but intimately coupled, contributions to the total activation barrier ($F_{\mathrm{tot}} = F_B + F_{\mathrm{el}}$): a local cage-scale barrier (F_B) associated with large-amplitude particle hopping, and a longer-range collective elastic barrier (F_{el}) associated with the need for facilitating small-amplitude elastic dilational displacements of all particles outside the cage to allow particle hopping. Recent ECNLE theory calculations for dynamical gradients near free surfaces have predicted that the near-field T_g gradient decays in an exponential manner as a function of distance (z) from the surface, but at large enough z decays as a scale-free inverse power law ($\sim z^{-1}$)[33]. The latter is a direct signature of the importance of collective elasticity for structural relaxation, with ECNLE theory predicting that the elastic barrier becomes increasingly dominant relative to its local cage analogue upon cooling to T_g (refs. [12,13]). This power-law tail decay form is a unique prediction, which, if validated, would provide new support for the idea that structural relaxation is of a coupled local–nonlocal character. After reporting our new simulation results, we quantitatively compare the data with a new asymptotic analytic ECNLE theory analysis and higher-precision numerical calculations for the spatial form of the T_g and α relaxation time gradients.

In Fig. 1a we report the simulated normalized T_g shift gradient as a function of distance z from the free surface of the film, where T_g is defined on a computational relaxation timescale of $10^5\ \tau_{LJ}$ (approximately 100 ns; τ_{LJ} is the Lennard Jones time unit as is of order 1 ns) based on purely in-equilibrium relaxation data (see Supplementary Fig. 2 for the underlying relaxation-time temperature dependences). As can be seen in Fig. 1a, the near-surface T_g gradient obeys an exponential form over the first 8–10σ (where σ is the Lennard Jones unit of length and is of order 1 nm) from the free surface, consistent with previous simulation studies of dynamics near vapour surfaces[26,34–36]. However, beyond $z \sim 10\sigma$, the gradient transitions to a long-range tail that has not previously been reported in simulation or experiment. Large-z upward deviations from the exponential decay are well beyond uncertainty (propagated from standard deviations over three film and three bulk runs) and are qualitatively robust to the choice of relaxation time versus temperature fitting form and timescale convention employed to extract T_g (Supplementary Figs. 3 and 4). Moreover, because the film data are normalized by data from very large bulk simulations, the large-z tail behaviour cannot be an anomaly of normalization against a quantity that encodes a residual interface effect. Figure 1a illustrates that the long-range tail of the T_g gradient can be described as a power law with a slope of −1, although other slowly decaying empirical descriptions are probably possible given the z-range of available data (which pushes the limit of current simulation capabilities) and their confidence intervals.

As can be seen in Supplementary Fig. 5, results for the α relaxation time gradient in $\tau_{\alpha,\mathrm{bulk}}/\tau_\alpha(z)$ also exhibit a two-regime form: a double-exponential form dominates for $z < 10\sigma$ (consistent with previous simulation and experimental findings over this length scale range[18,26,37–39]), followed by a longer-ranged tail. The form of this tail is further clarified by fitting[40] to data at multiple temperatures to obtain a reduced-noise calculation of the relaxation time at low temperature. As seen in Fig. 1b, the long-range tail of $\log(\tau_{\alpha,\mathrm{bulk}}/\tau_\alpha(z))$ can similarly be described by an inverse power law of essentially an identical form as found for the T_g gradient.

These simulation data are inconsistent with the gradient forms predicted by multiple theories of glass formation. RFOT predicts that the difference between the film and bulk relaxation rates decays exponentially with distance from the surface[29]; this conflicts with the low-z regime, where empirically the relaxation time decays in a double-exponential manner. RFOT also does not predict any long-range tail, and, as shown in Supplementary Fig. 6, a multi-parameter fit of its predicted dynamic gradient form[29] does not qualitatively describe the simulation data in either regime. The cooperative free volume theory—a very different model—predicts a similar exponential relaxation time gradient[31] that does not agree with either regime observed in simulation. Alternative phenomenological free-volume-based slow domain percolation models predict a purely power-law T_g gradient[8] and do not qualitatively capture the short-ranged exponential decay that quantitatively dominates the near-surface behaviour. The phenomenological 'cooperative string' theory predicts a roughly double-exponential decay of relaxation times, but does not capture the long-range power-law tail[32]. Broadly, the far-field gradient appears to be qualitatively inconsistent with theories based on a static or dynamic length scale that is cut off by a boundary in a critical phenomena spirit. This observation is strongly buttressed by previous results indicating a saturation (not an unbounded growth) in the range of surface gradients (or intrinsic dynamic length scale) on cooling[26,37,41,42].

Instead, our finding of a short-range, large-amplitude (double) exponential gradient of (α time) T_g followed by a lower-amplitude tail suggests two distinct physical mechanisms for dynamical gradients—one medium-ranged and one long-ranged. The presence of two mechanisms for the gradient is then suggestive of two coexisting physical effects driving slow dynamics in the bulk fluid. If we specifically consider that an inverse power law describes our long-range tail data, such

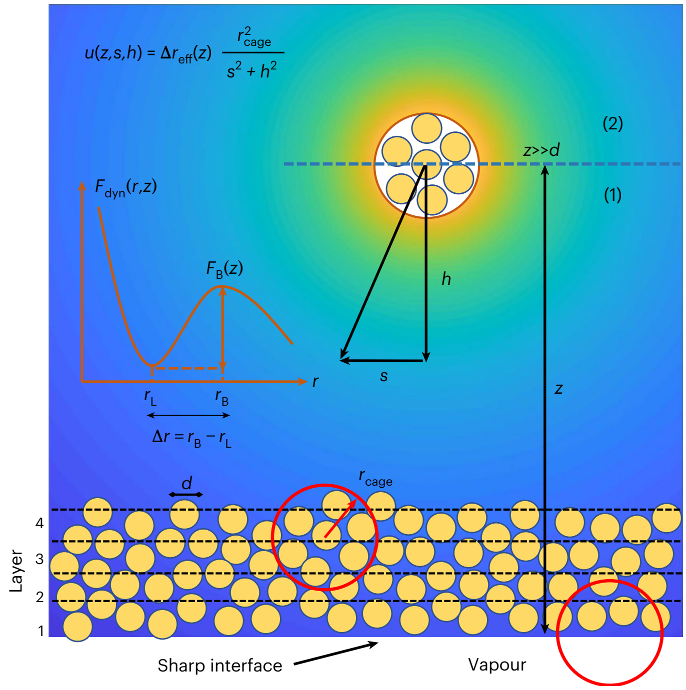

Fig. 2 | Schematic of the film and key concepts of ECNLE theory, where 'layers' are shown only for illustrative reasons and density is homogeneous in all directions within the film. Regions (1) and (2) correspond to the lower and upper half space, respectively, of a hopping event characterized by the centre of the cage (radius r_{cage}) located at z. The corresponding collective elastic displacement field (u) of all particles outside the cage has an amplitude set by the effective jump distance (Δr_{eff}) and depends on the defined distance variables z, h and s. The background is coloured as the logarithm of $r_{\mathrm{cage}}^2/(s^2+h^2)$ and reports on the local elastic displacement absent the gradient in $\Delta r_{\mathrm{eff}}(z)$, with more yellow colours denoting larger displacements. The left inset in red provides a schematic illustration of the dynamic free energy as a function of tagged particle (diameter, d) displacement, r, in a liquid of density ρ. The localization length (r_L), barrier location (r_B), jump distance (Δr) and local cage barrier (F_B) are defined. More details of the model and theory are available in ref. 33 and the Supplementary Information.

behaviour is potentially consistent with elastic activation theories that posit that much of the dynamic activation barrier in sufficiently supercooled fragile liquids resides in a scale-free long-range collective elastic field[10].

More broadly, this type of two-mechanism scenario is the hallmark of the ECNLE theory, based on its two causally coupled barriers, as discussed above. The fundamental quantity in this theory is the dynamic free energy that controls particle trajectories (Fig. 2), which is constructed from knowledge of the pair structural correlations, and from which both barriers and the α relaxation time can be predicted. ECNLE theory has been shown to capture well, often with no adjustable parameters, the temperature and density dependences of the relaxation time of diverse bulk glass-forming liquids and suspensions, including hard spheres, colloids, small molecules and polymers[12,13,43], where 'cooperativity' and the non-Arrhenius nature of the α relaxation time are dominated by the collective elastic barrier.

In the symmetry-broken film (Fig. 2), both barriers in ECNLE theory depend on distance from the free surface via the z-dependence of the dynamic free energy, solely for kinetic reasons not related to any local gradient in structure or thermodynamics induced by an interface[33]. Rather, the interface enters simply as a step function separating a vapour of zero density from a liquid of bulk density, ρ. The spatially heterogeneous relaxation time gradient is a consequence of[33,44] (1) softening of dynamic cage-scale constraints nucleated at the surface due to loss of nearest neighbours, which are transferred into the film via a layer-by-layer facilitation-like process that fully controls the local cage barrier gradient and (2) a gradient in the elastic barrier driven by three effects: a z-dependent amplitude of the elastic displacement field (set by the effective jump distance, Δr_{eff}), a z-dependent local elastic modulus or harmonic spring constant of the localized state (K_0), which varies as the inverse square of the dynamic localization length r_L, and geometric truncation of the elastic displacement field at the vapour interface. The local z-dependent quantities Δr_{eff} and K_0, which grow with cooling or densification, decay exponentially towards their bulk values with increasing distance from the interface with a characteristic length scale of a few particle diameters[44]. Previous work focused on dynamic gradient predictions within the first approximately ten particle diameters of the interface, where the theory has been shown to correctly predict the exponential decay of $T_g(z)$ and the double-exponential decay of the α time in the immediate vicinity of the interface, as well as spatially inhomogeneous decoupling[37], and factorization of the spatial and temperature dependences of the total barrier[26,33]. However, at large enough distances from the interface where F_B, Δr_{eff} and K_0 all attain their bulk values, the elastic barrier still varies with increasing z, but only due to the cutoff of the displacement field at the vapour interface. This is the physical origin of the long-range inverse power-law decay of dynamic gradients of T_g and α time predicted by the theory, as briefly discussed previously[33].

To better understand the predictions of ECNLE theory for the long-range tail of the dynamic gradients, we have carried out a new analytic analysis (with prefactors) using the same foundational hard-sphere fluid model employed previously[33]. As discussed above (and Fig. 2), for a relaxation event far from the interface ($z \gg d$), the total barrier is spatially varying only due to the elastic displacement field cutoff effect. The theory predicts that the maximum amplitude of the displacement field is sufficiently small that a linear elastic analysis of its form, and hence a harmonic calculation of the corresponding elastic barrier, is valid[12]. Thus, summing the elastic energy of all particles outside the cage[33] (Fig. 2 and Supplementary Information provide details), the contribution to the elastic barrier from the upper half space is simply half of its bulk value

$$\begin{aligned} F_{\mathrm{el}}^{(2)} &= 2\pi\rho \frac{K_0 \Delta r_{\mathrm{eff}}^2}{2} r_{\mathrm{cage}}^4 \\ &\times \left[\int_0^{r_{\mathrm{cage}}} \mathrm{d}h \int_{\sqrt{r_{\mathrm{cage}}^2 - h^2}}^{\infty} \frac{s\mathrm{d}s}{(s^2+h^2)^2} + \int_{r_{\mathrm{cage}}}^{\infty} \mathrm{d}h \int_0^{\infty} \frac{s\mathrm{d}s}{(s^2+h^2)^2} \right] \\ &= \frac{F_{\mathrm{el}}^{\mathrm{bulk}}}{2} \end{aligned} \quad (1)$$

whereas the contribution from the lower-half space is reduced by the interface as

$$\begin{aligned} F_{\mathrm{el}}^{(1)} &= 2\pi\rho \frac{K_0 \Delta r_{\mathrm{eff}}^2}{2} r_{\mathrm{cage}}^4 \\ &\times \left[\int_0^{r_{\mathrm{cage}}} \mathrm{d}h \int_{\sqrt{r_{\mathrm{cage}}^2 - h^2}}^{\infty} \frac{s\mathrm{d}s}{(s^2+h^2)^2} + \int_{r_{\mathrm{cage}}}^{z} \mathrm{d}h \int_0^{\infty} \frac{s\mathrm{d}s}{(s^2+h^2)^2} \right] \\ &= \frac{F_{\mathrm{el}}^{\mathrm{bulk}}}{2} \left[1 - \frac{r_{\mathrm{cage}}}{2z} \right] \end{aligned} \quad (2)$$

where $F_{\mathrm{el}}^{\mathrm{bulk}}$ is the bulk elastic barrier, r_{cage} is the cage size determined from the bulk pair correlation function $g(r)$, and the geometry variables s and h are defined in Fig. 2. The subtractive term in the final equality of equation (2) varies as $1/z$ and is a direct consequence of the cutoff of the elastic displacement field at a planar interface. Combining equations (1) and (2) yields

$$1 - \frac{F_{\mathrm{el}}(z)}{F_{\mathrm{el}}^{\mathrm{bulk}}} = \frac{r_{\mathrm{cage}}}{4z}, \quad z \gg d \quad (3)$$

Per standard ECNLE theory analysis[12,33], the mean α relaxation time is identified with the mean Kramer's time to cross the barrier of the z-dependent dynamic free energy including elastic barrier physics. At large z this leads analytically (Supplementary Information) to the conclusion that both the logarithm of the mean α time and the relative deviation in T_g obey the same inverse power law for $z \gg d$:

$$\frac{T_{\mathrm{g,bulk}} - T_g(z)}{T_{\mathrm{g,bulk}}} \approx \frac{F_{\mathrm{el}}^{\mathrm{bulk}}}{F_B^{\mathrm{bulk}} + F_{\mathrm{el}}^{\mathrm{bulk}}} \frac{r_{\mathrm{cage}}}{4z}. \quad (4)$$

Given that previous ECNLE theory studies have shown numerically and predicted analytically[33] (well-verified in simulation[26,39]) that the near-field gradients in the T_g deviation and $\log(\tau_\alpha(z)/\tau_{\alpha,\mathrm{bulk}})$ are exponential, we can now propose convenient approximate analytic forms for the full gradients by simply summing the predicted two asymptotic (near and far from the interface) functional forms to obtain

$$\frac{T_{\mathrm{g,bulk}} - T_g}{T_{\mathrm{g,bulk}}} \approx A \exp\left(-\frac{z}{\xi}\right) + \frac{B}{z}, \quad (5)$$

$$\log\left(\frac{\tau_{\alpha,\mathrm{bulk}}}{\tau_\alpha(z)}\right) \approx a \exp\left(-\frac{z}{\xi_\tau}\right) + \frac{b}{z}. \quad (6)$$

Equations (5) and (6) do not represent the full ECNLE theory numerical calculations; rather, they are inspired by the ECNLE analytic asymptotics, which we can employ to analyse our simulation data. As seen in Fig. 1a, this form provides an excellent fit to the simulated T_g gradient data over the full z-range probed (with A, B and ξ for the simulated $T_g(z)$ gradients equal to 0.20, 0.034 and 2.6σ, respectively).

To buttress the above asymptotic analytic analysis, we perform new, more numerically accurate, ECNLE theory calculations of the long-range gradient and other key dynamical properties defined in Fig. 2 (see Fig. 3). These involve no fitting parameters and exhibit (as expected) a modestly more complex crossover between the low-z and high-z asymptotic regimes than described by equations (5) and (6) (implementation details are provided in Methods and Supplementary Information). We emphasize that no physical adjustments have been made to the prior theory in these new calculations to accommodate the new simulation data above. Instead, they involve only more precise numerical computation of the long-range tail and elastic barrier. Figure 1c,d explicitly shows that the numerical ECNLE theory results capture the simulation findings for both the $T_g(z)$ and $\tau_\alpha(z)$ dynamic gradients: a short-ranged high-amplitude (double) exponential gradient for ($\tau_\alpha(z)$) $T_g(z)$ is predicted to cross over to a weak power-law decay for large z. Within the modest difference in the elementary length scales d and σ (Methods), theory and simulation are in semi-quantitative agreement for the key features of the decay: the crossover distance from exponential to power law (approximately ten segmental or bead diameters), the fractional recovery of bulk-like T_g at this crossover (~1.5 order of magnitude drop in T_g perturbation from the free surface), and the slowly decaying tail in T_g. It is notable that this agreement is found despite the fact that the theory is implemented for the simple hard-sphere model while the simulations are of a bead-spring polymer, thus emphasizing the semi-universality of these gradient effects. The two-regime nature of the T_g gradient, its initial exponential decay, and the power-law form and slope of the long-range tail are unambiguous signatures of the ECNLE theory physics: interface-perturbed medium-range softening of cage constraints (affects both barriers) and a longer-ranged geometric cutoff of the associated elastic displacement field. The good numerical agreement between theory and simulation provides further support for the basic physical ideas of ECNLE theory under broken-symmetry film conditions.

A natural question is whether the behaviour described above may qualitatively change at timescales appreciably longer than those

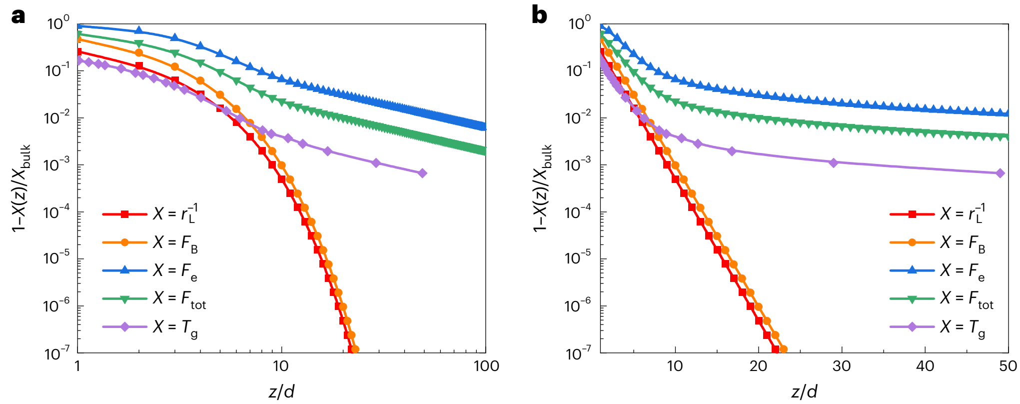

Fig. 3 | Theoretical predictions for the fractional and normalized deviations from bulk of multiple dynamical properties as a function of distance from the vapour interface. a,**b**, ECNLE theory predictions of various (normalized by the bulk) dynamic quantities as a function of dimensionless distance from the interface on logarithmic (**a**) and linear (**b**) distance scales. Results are shown for the inverse dynamic localization length $1/r_L$, local cage barrier F_B, elastic barrier F_e, total barrier $F_{tot} = F_B + F_{el}$ and T_g.

accessible to simulations. In this regard, we recall that previous simulation, experimental and ECNLE theory studies have all indicated that local α relaxation times in films obey a fractional power-law decoupling relation with the bulk analogue[26,33,37,45], an important consequence of which is that T_g gradients deduced from equilibrium dynamics are relatively timescale-insensitive[26]. Indeed, as shown in Supplementary Figs. 3 and 4, when the timescale convention for determination of T_g from the simulations is varied by two decades, the qualitative behaviour is preserved, with a weak trend towards stronger power-law tails at lower temperatures. ECNLE theory extends the analysis of this question to much longer experimental timescales. As shown in Fig. 1c, the predicted T_g gradient varies only weakly as the timescale convention is changed over an enormous range from simulation to experimental timescales, and in a direction consistent with simulation. Recent experimental evidence confirms that the near-interface double-exponential relaxation time gradient persists until at least near the experimental-timescale T_g determined from a laboratory 100-s vitrification criterion, with an intrinsic dynamic decay length comparable to that reported here from simulation and theory[18]. Although those experiments did not probe sufficiently deeply into the film to access the regime where the power-law tail reported here is expected, combined with our simulations the experiments do set bounds on the tail behaviour. At simulation timescales, the power law is relatively low in magnitude and probably grows weakly on cooling; however, it is not so strong at the experimental T_g as to overwhelm the near-surface exponential (double-exponential) gradient of T_g (relaxation time).

The experimental impacts of the long-range tail we report will be difficult to resolve in most mean-film-property measurements. The most sensitive of these measurements have provided some insight into the distribution of relaxation times within the film, such as dielectric spectroscopy measurements reporting on broadening of the mean relaxation process in films relative to bulk[46]. These experiments point to the presence of a domain of faster dynamics near the surface, but the observed broadening is dominated by the short-range double-exponential gradient in τ_α. The effect of the long-range contribution to the gradient would be within the extreme tail of this film-averaged distribution, and whether the long-range gradient tail could be extracted or inferred from film-averaged data will require future quantitative theoretical analysis. We believe the experimental prospects are challenging given practical uncertainties and convolution with the bulk breadth of the relaxation process. Similarly, experiments that employ optical photobleaching have pointed to the presence of two relaxation processes in the film below T_g, with one attributed to surface relaxation and one to bulk-like relaxation[47]. This behaviour (and also broadening of the film-averaged frequency-domain dielectric loss) has been previously predicted by the ECNLE theory[48]. However, it is dominated within the theory by the steep near-surface double-exponential gradient of the α relaxation time and is probably not a good target for experimental observation of the long-ranged tail. One possible whole-film target for observation of these effects may be in intermediate-thickness films, where the spatially long-range tail can compensate for its relatively low-magnitude, resulting in an appreciable contribution to the overall shift in relaxation time or T_g. Indeed, this long-range tail is probably responsible for our recent simulation report of excess accelerations in dynamics in intermediate films relative to what would be expected from exponential gradients alone[39].

Concerning the ECNLE theory predictions on the 100-s timescale of the T_g gradient in Fig. 1d, the quantitative results suggest that direct spatially resolved experimental observation of the power-law tail in T_g would require sub-kelvin sensitivity to T_g shifts, which would probably be extraordinarily challenging in conjunction with the demand for spatial resolution. More promisingly, observing the long-range tail in the relaxation-time spatial gradient would require sufficient sensitivity to detect timescale shifts of order 10–60%—relatively tractable from a sensitivity standpoint. However, the demand that these measurements be spatially resolved out to distances of several tens of nanometres from the interface will require new advances in spatially resolved measurements near interfaces to extend further into the material[18,49,50].

Importantly, ECNLE theory also predicts how the amplitude of the power-law tail varies with chemistry, and for the same physical reason it predicts a modest dependence on the vitrification timescale criterion. Within the theory, the magnitude of collective elastic relative to local cage barriers at T_g is non-universal, and directly related to the bulk dynamic fragility[51] (Supplementary Information). Liquids with a weaker relative contribution from collective elasticity (for example, strong glass-formers) are predicted to exhibit a weaker long-range power-law tail in their interfacial dynamics gradient (Supplementary Information). This provides an additional testable prediction for future simulations and experiments possessing sufficient sensitivity to resolve the dynamic gradient tail. The question of the dependence of perturbed surface dynamics on materials chemistry has long been a focus of the field for both practical and fundamental reasons, and very recent experimental results have indeed suggested a strong dependence of these perturbations on the bulk-liquid dynamic fragility[52,53].

To conclude, the discovery of a long-range, apparently scale-free dynamical gradient induced by a single interface suggests that low-magnitude interfacial perturbations to T_g and dynamics in nanostructured materials can persist over a remarkably long-range. Indeed, a $1/z$ power-law tail in $T_g(z)$and $\tau_\alpha(z)$ formally has an infinite range (divergent zeroth and first moments), which, based on our theoretical analysis, is not expected to change at an ultra large distance from the interface. Our findings may provide a theoretical starting point for understanding the puzzling observation of exceptionally long-range interfacial perturbations of T_g in a variety of experimental systems, including polymeric bilayer films[54,55]. Fundamentally, our core simulation findings provide a new direct validation of a signature foundational prediction of ECNLE theory—that collective elasticity plays a critical role in structural relaxation in deeply supercooled liquids. This provides further support for this approach, which has successfully addressed other diverse glassy phenomena, including nanoconfinement effects on equilibrium dynamics over a large range of film thicknesses[39], the temperature and density dependences of the bulk α relaxation time in a large range of polymers[43], small molecules[13] and colloidal glass-formers[12], the emergence of diffusion-relaxation decoupling in deeply supercooled liquids[51], and a non-causal relationship between thermodynamics and dynamics in molecular, polymer and inorganic glass-forming liquids[56].

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-023-01995-8.

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Methods

Details of the simulations, which employ a bead-spring polymer model[57] modified to exhibit exceptional crystallization resistance[58], are provided in the Supplementary Information. Simulated films, which are ~50 segmental (bead) diameters (σ) thick (thus of order 50 nm thick in real units) and comparably wide (~$40 \times 40\sigma$), are supported on a nearly dynamically neutral substrate such that dynamical perturbations in the film are dominated only by a single free surface. Films are supported on a substrate with a substrate–polymer pairwise interaction strength that is tuned to be 51.1% of the polymer–polymer bead interaction strength, which yields almost no local gradient in dynamics (Supplementary Fig. 1). As compared to the large literature of molecular simulations of two-interface free-standing films[16,26,59], this allows for quantification of altered dynamics much further from the dynamically active free surface without interference by a second surface gradient—up to at least 35σ (~35 nm) from the free surface. Corresponding bulk relaxation times at the lowest temperatures simulated approach 10^5 LJ time units τ_{LJ} (~100 ns). We compute $T_g(z)$ by fitting the temperature dependence of the relaxation time $\tau_\alpha(z)$ at each position z and then defining their $T_g(z)$ as the temperature at which the local relaxation time is 10^5 LJ time units. We perform this fit using two very different widely employed empirical functional forms—the Vogel–Fulcher Tammann equation[60,61] (results shown in the Supplementary Information) and the 'cooperative model' of Schmidtke et al.[40]—to ensure that the results are not a consequence of the choice of fit form. Three replicas of this simulation and of the corresponding bulk system are simulated, and error bars are propagated from these replicas. Together with averaging over multiple runs, this protocol provides sufficient range and sufficient statistical power to resolve the long-range form of the gradient in T_g emanating from a free surface.

To numerically compute $T_g(z)$ gradients from ECNLE theory, we adopt a well-established method[12,37] to map the hard-sphere model to a typical fragile liquid (polystyrene)[43] at 1 atm (additional details are provided in the Supplementary Information). We also present numerical relaxation time-gradient calculations that are not tied to the polystyrene mapping. For the simulated flexible polymer bead-spring model, the elementary length scale σ is expected to be of order, but modestly smaller than, the hard-sphere theory diameter d based on the mapping procedure for polymers[43] and earlier simulation-theory studies[39].

Data availability

All relevant data are included in the paper and/or its Supplementary Information files. Raw simulation trajectory files, which are prohibitively large, are available upon reasonable request from D.S.S.

Code availability

All results in this paper employed openly available codes and/or standard numerical algorithms.

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Author contributions

The paper and Supplementary Information were written based on the contributions of all authors. A.G. performed all simulations under the supervision of D.S.S. A.G. and D.S.S. jointly conceived of and analysed all simulations. A.D.P. and K.S.S. performed all ECNLE theory analytical analysis and numerical calculations.

Competing interests

The authors declare no competing interests.

Additional information

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Signature of collective elastic glass physics in surface-induced long-range tails in dynamical gradients

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Supplementary Information for "Signature of collective elastic glass physics in surface-induced long-range tails in dynamical gradients"

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Detailed simulation methods

Simulations employ a variant of the bead-spring model of Kremer and Grest[1] that includes attractive interactions between non-bonded beads. This variant employs a reduced backbone bond length in order to enhance crystallization resistance in the presence of an ordered substrate[2]. Within this model, non-bonded beads interact via the 12-6 Lennard-Jones (LJ) potential,

$$E_{ij} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] \tag{1}$$

with the pair interaction truncated beyond r_{cut} = 2.5. Nonbonded polymer beads interact pairwise via this form with ε = 1 and σ = 1 σ_{LJ} (where σ_{LJ} is the Lennard Jones unit of length and corresponds to approximately 1 nm in real units[3–5]). Bonded beads interact via the Finitely Extensible Nonlinear Elastic (FENE) potential superposed with the 12-6 LJ interaction,

$$E_{FENE} = -0.5KR_0^2 \ln\left[1-\left(\frac{r}{R_0}\right)^2\right] + 4\varepsilon_{FENE}\left[\left(\frac{\sigma_{FENE}}{r}\right)^{12} - \left(\frac{\sigma_{FENE}}{r}\right)^{6}\right] + \varepsilon_{FENE} \tag{2}$$

where K = 30, R_0 = 1.3, ε_{FENE} = 1, σ_{FENE} = 0.8, and where the LJ term is truncated at its minimum of $2^{1/6}\,\sigma_{FENE}$.

We simulate 5000 20-bead chains supported on a crystalline substrate. The substrate is comprised of an FCC lattice of LJ beads at a reduced density of 1.4, with the [111] face exposed to the polymer. Polymer and wall beads interact via equation (1), albeit with σ = 1 and $\varepsilon_{wall\text{-}polymer}$ = 0.515. As shown below, this leads to a substrate that is dynamically neutral to a good approximation. The area of the surface is 41.7 σ_{LJ} *40 σ_{LJ}, and the polymer film fully coats the surface with a temperature-dependent thickness in the range of 63 to 48 σ_{LJ}. The upper surface of the polymer film is exposed to vacuum such that it is under zero pressure. Periodic boundary conditions are employed in the in-plane directions.

Simulations are performed in the LAMMPS molecular dynamics package[6]. Temperature is controlled via the Nose-Hoover thermostat, with a damping parameter of 2 τ_{LJ} (where τ_{LJ} is the LJ unit of time and corresponds to approximately 1 ps in real units). Simulations employ a timestep of 0.005 and use a Verlet integrator. No barostat is employed for the film simulations, but the presence of a free surface yields a system that obeys constant temperature, pressure, number (NPT) conditions at a pressure of P = 0.

Corresponding bulk simulations employ periodic boundary conditions in all directions and the Nose-Hoover thermostat/barostat pair, as implemented in LAMMPS with damping parameters of 2 τ_{LJ} and at a pressure of P=0. Bulk simulations likewise employ 5000 20-bead chains to ensure direct comparability of results at comparable size.

Systems are subjected to a thermal quench procedure defined by the Predictive Stepwise Quench (PreSQ) algorithm, which has been well-validated to yield in-equilibrium configurations for this model system[7]. In summary, an iterative thermal quench is employed with long thermal anneals at each temperature sufficient to ensure equilibrium segmental packing.

Both bulk and film systems are independently simulated 3 times, and all results reflect an average over these runs with error bars propagated from uncertainty over both film and bulk runs.

Relaxation times are quantified via the self-part of the intermediate scattering function, computed at a wavenumber of 7.07/σ_{LJ}, which corresponds approximately to the first peak in the segmental static structure factor. Relaxation times are extracted by fitting the time dependence of this function, at times beyond the initial picosecond relaxation process, to a stretched exponential fit for the purposes of smoothing and interpolation. The relaxation time is then defined as the time at which this function decays to a value of 0.2, consistent with a large number of prior simulation works[8–10].

These relaxation times are computed at a local level by first binning particles into bins of thickness 0.875 σ_{LJ} and then performing the analysis described above. Glass transition temperatures in the main text (Figure 1a) and Figure S 1 below are then obtained by fitting the temperature dependence of these data (see Figure S 2 below) to the empirical Cooperative Model of Schmidtke and coworkers[11] that accurately describes experimental alpha time data of fragile molecular and polymer glass forming liquids. These fits are employed to obtain T_g values at a timescale of 10^5 τ_{LJ}, which is at the upper limit of timescales simulated for the corresponding bulk system and thus in the middle of the simulated film. Additional results below are reported based on alternatively fitting to the Vogel-Fulcher-Tammann model and T_g is determined based on that fit to establish that results are qualitatively insensitive to the choice of fit model. In addition, the Cooperative Model fit across the full temperature range is used produce the relaxation time gradients shown in Figure 1b of the main paper, which involve less noise than the corresponding single-temperature data shown in Figure S 5 below.

In comparing our results to the RFOT theory, we employ the equation derived by Stevenson and Wolynes[12] for the spatial variation of the relaxation time near a free surface:

$$\tau_\alpha(z)^{-1} \approx \left(\tau_{\alpha,surf}^{-1} - \tau_{\alpha,bulk}^{-1}\right)\exp\left(-\frac{z}{\xi_{MCT}}\right) + \tau_{\alpha,bulk}^{-1}\,. \tag{3}$$

We rearrange this equation to obtain the logarithmic ratio of bulk to local relaxation time as predicted by RFOT:

$$\log\frac{\tau_{\alpha,bulk}}{\tau_\alpha(z)} = \log\left\{A\exp\left[-\frac{z}{\xi_{MCT}}\right]+1\right\} \tag{4}$$

where

$$A=\left(\frac{\tau_{\alpha,bulk}}{\tau_{\alpha,surface}}-1\right) \tag{5}$$

To assess the best possible description that the RFOT form could provide of the data, we then empirically fit the parameters A and ξ_{MCT} to the low-z regime of data.

The resulting two-parameter fit of SI equation (4) is shown in Figure S6. As can be seen in panel (a), equation (4) does not predict the long-range power law tail observed in the simulation data for $\log(\tau_{\alpha,bulk}/\tau_{\alpha}(z))$. Moreover, as emphasized in Figure S6 panels (b), (c), and (d) which zoom in on the low-z regime, the RFOT functional form also does not correctly describe the variation in $\tau_{\alpha}(z)$ near the interface. Instead, there is a systematic and qualitative deviation well beyond uncertainty, with the 2-parameter fit of equation (4) underpredicting the relaxation time acceleration at the surface, overpredicting it for approximately $3 < z < 7$, and again underpredicting it at higher z. Panels (c) and (d) make clear that in the near surface lower z regime the simulation data are described by a double exponential decay of the alpha time back to bulk.

Supplementary simulation data

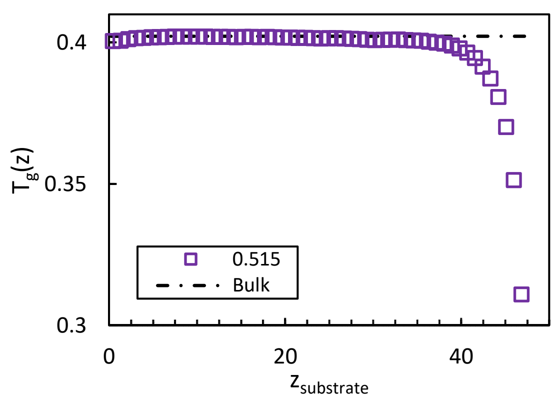

Figure S 1. Glass transition temperature at a vitrification timescale of 10^5 τ_{LJ} vs dimensionless distance (in dimensionless LJ distance units σ) from the substrate for $\varepsilon_{wall\text{-}polymer}$= 0.515, the substrate interaction employed in the study. The results demonstrate the near dynamic neutrality of the substrate.

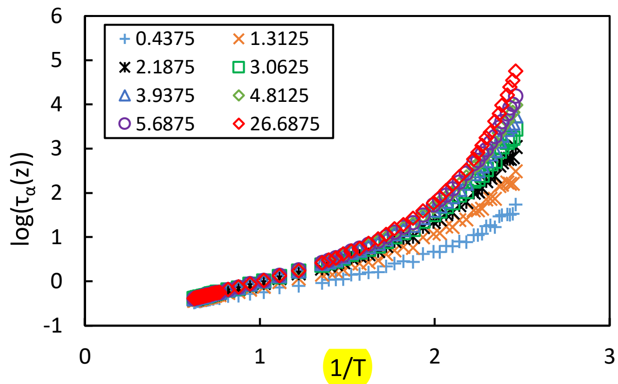

Figure S 2. Segmental relaxation times vs inverse temperature for layers of segments at central distances z from the free surface of 0.4375 σ_{LJ}, 1.3125 σ_{LJ}, 2.1875 σ_{LJ}, 3.0625 σ_{LJ}, 3.9375 σ_{LJ}, 4.8125 σ_{LJ}, 5.6875 σ_{LJ}, and 26.6875 σ_{LJ}

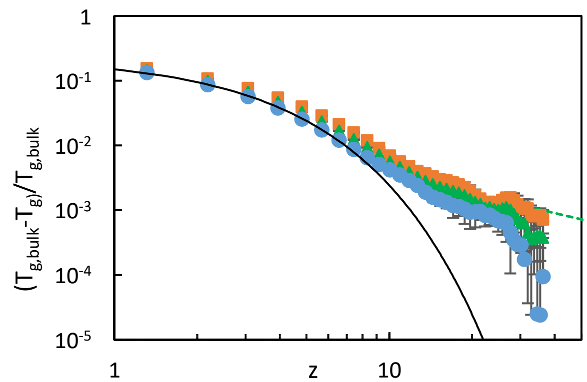

Figure S 3. Normalized glass transition temperature gradients (as computed via fits of the Cooperative Model of Schmidtke and coworkers[11] to our simulation data as discussed above) using vitrification time scales of 10^3 τ_{LJ} (blue circles), 10^4 τ_{LJ} (green triangles), and 10^5 τ_{LJ} (orange squares). The black curve corresponds to an exponential fit to the low-z data at 10^5 τ_{LJ}. The straight line is a power law decay with exponent -1. Data are presented as mean values with error bars reflecting standard deviations propagated from data for 3 films and 3 bulk runs.

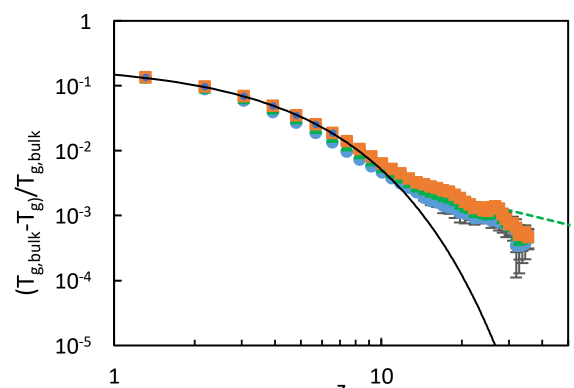

Figure S 4. Normalized glass transition temperature gradients (as computed via fits of the Vogel Fulcher Tammann[13,14] form to our simulation data) as a function of distance from the free surface computed using vitrification time scales of 10^3 τ_{LJ} (blue circles), 10^4 τ_{LJ} (green triangles), and 10^5 τ_{LJ} (orange squares). The black curve corresponds to an exponential fit to the low-z data at 10^5 τ_{LJ}. The straight line is a power law decay with exponent -1. Data are presented as mean values with error bars reflecting standard deviations propagated from data for 3 films and 3 bulk runs.

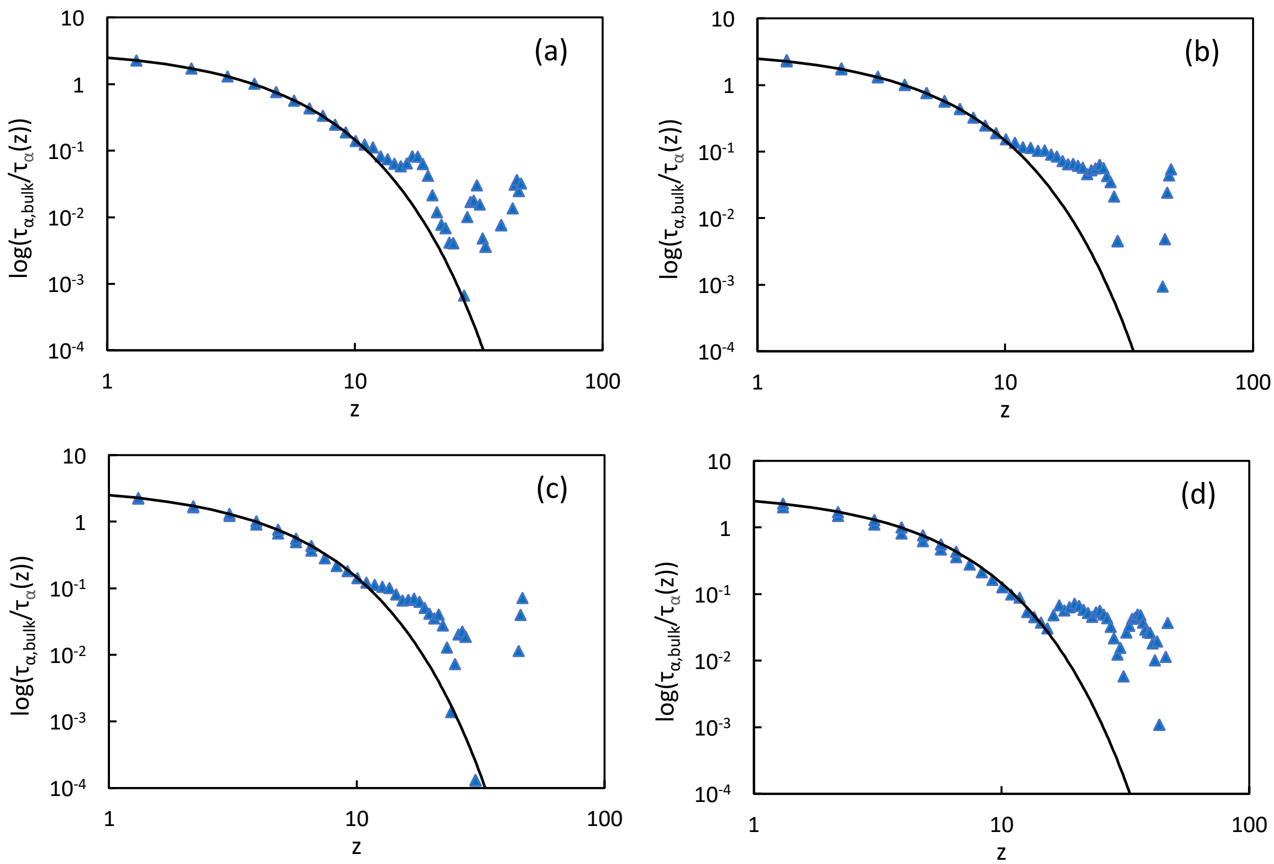

Figure S 5. Simulated segmental relaxation time normalized by bulk relaxation time versus normalized distance from the free surface at the four lowest temperatures simulated: reduced LJ temperatures of (a) 0.406, (b) 0.409, (c) 0.411, and (d) 0.414. Black lines are exponential fits to the low-z data.

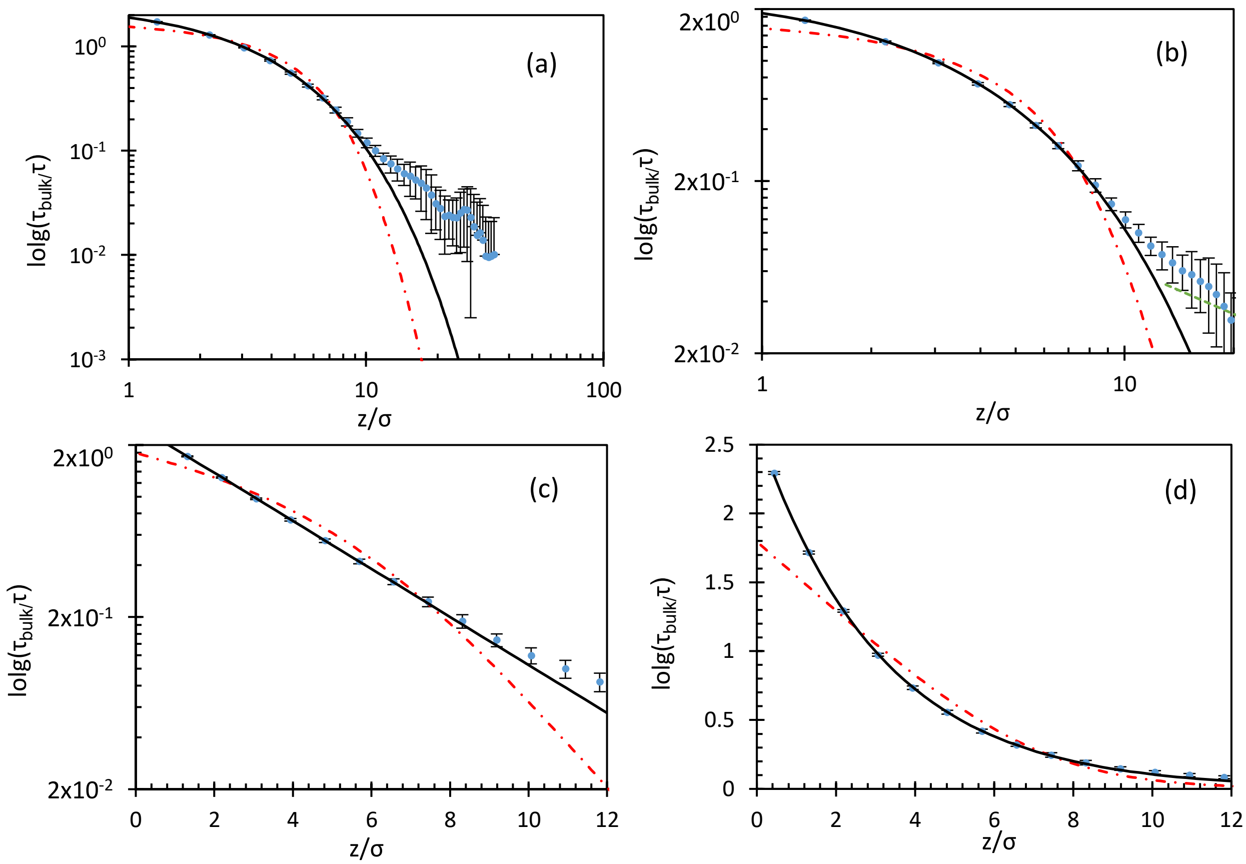

Figure S 6. (a) Simulated relaxation time gradients as in main paper Figure 1b (points and associated confidence intervals). Solid black lines are fits of low-z data to a decaying exponential. Dot-dashed red curve is a two-parameter free fit to the gradient form predicted by the Random First Order Transition Theory[12] (equation (4) in the SI). Panel (b) shows same data, but focused on low-z region. Panel (c) shows same data, but focused on low-z domain and with a linear abscissa. Panel (d) shows the same data, focused on low-z, on a linear-linear plot. In all panels, data are presented as mean values with error bars reflecting standard deviations propagated from data for 3 films and 3 bulk runs.

Spatially long range tail of the glass transition temperature and alpha time gradients in ECNLE theory

As background for the reader, we first sketch elements of the published ECNLE theory[15–17] focusing on the thick film with one vapor interface system[18,19]. We then perform a detailed analytic analysis that was only briefly sketched in prior work[18] and sketched in the main article for the effect of long range collective elasticity on the form of the spatial gradient of the glass transition temperature and the alpha time *far* from the interface. We emphasize that the mathematical simplifications invoked below are *not* made in our numerical calculations reported in the main text, nor in the earlier theoretical work where all details are explained[18]. Rather, these simplifications are employed here solely to derive analytic results we believe do capture the numerical predictions of the theory far from the interface. The fact that predictions based on this analysis agrees well with our full numerical calculations strongly supports the simplifications adopted.

We first briefly summarize the ECNLE theory of a thick film with one vapor sharp (step function) interface developed previously[18]. The core ideas of the vapor-interface-induced softening and spatial gradient of the dynamic free energy are illustrated in Figure 2 of the main manuscript for the foundational hard-sphere fluid model characterized by the particle diameter, d, and bulk fluid density number, ρ, or corresponding volume fraction, $\Phi = \frac{\rho \pi d^3}{6}$. The minimalist theory ignores any equilibrium spatial gradients of density or changes of the isotropic averaged pair structure as quantified by radial distribution function g(r) or static structure factor, S(q). The dynamic free energy describes motion of a tagged particle within a cage formed by its neighbors, and for a one-interface thick film is constructed based on a facilitation perspective of how reduced caging constraints at a vapor interface (due to loss of nearest neighbors) is spatially transferred into the film interior as a function of distance z from the vapor interface (or at layer *i*)[18,19]:

$$F_{dyn}(r,z) = F_{ideal}(r) + F_{caging}^{bulk}(r)\left(1 - \frac{1}{2^{(z+d)/d}}\right) = F_{dyn}^{bulk}(r) - \frac{F_{caging}^{bulk}(r)}{2^i} \qquad (6)$$

where $z = (i-1)d$, r is the displacement of a tagged particle from its initial position, $F_{ideal}(r) = -3k_B T \ln\left(\frac{r}{d}\right)$ is the ideal entropy like contribution, k_B is Boltzmann's constant, and T is temperature. The local caging component of the dynamic free energy in the bulk is:

$$F_{caging}^{bulk}(r) = -\rho k_B T \int \frac{d\mathbf{q}}{(2\pi)^3} \frac{S(q)C^2(q)}{1+S^{-1}(q)} \exp\left[-\frac{q^2 r^2}{6}\left(1+S^{-1}(q)\right)\right] \qquad (7)$$

where q is the wavevector, $C(q) = \left[1 - S^{-1}(q)\right]/\rho$ is the direct correlation function, and $S(q)$ is calculated using Percus-Yevick integral equation theory of the hard sphere fluid. The key local length and energy scales of the dynamic free energy are the localization length, r_L, barrier position, r_B, jump distance, $\Delta r = r_B - r_L$, and local cage barrier, $F_B(z) = F_{dyn}(r_B, z) - F_{dyn}(r_L, z)$, as depicted in Figure 2 of the main manuscript. Far from the interface (z >> d), the dynamic free energy recovers its bulk value $F_{dyn}(r,z) \to F_{dyn}^{bulk}(r)$.

Phenomenological "elastic models" (e.g., the shoving model)[20,21] suggest a strong and mechanistic correlation between large amplitude local hopping and a longer-range collective elastic physics. In ECNLE theory, cage escape is argued to require a small cage expansion which induces a small harmonic displacement field involving all particles outside the cage. Details of the physical and mathematical treatment of the surface-modified displacement field are given in reference[18]. When considering an activated hopping event deep in the film far from the interface, the displacement field in the lower half space (region (1) as depicted in Figure 2 in the main manuscript) is per a bulk system. The form of the displacement field also recovers its bulk form and is given by

$$u(s,h) = \Delta r_{eff} \frac{r_{cage}^2}{h^2 + s^2} \tag{8}$$

where r_{cage} is the particle cage radius determined as the location of the first minimum of $g(r)$, the variables s and h are defined in Figure 2 of the main manuscript, and Δr_{eff} is the amplitude of the displacement field given by:

$$\Delta r_{eff} = \frac{3}{r_{cage}^3}\left(\frac{\Delta r^2 r_{cage}^2}{32} - \frac{\Delta r^3 r_{cage}}{192} + \frac{\Delta r^4}{3072}\right) \approx \frac{3}{32}\frac{\Delta r^2}{r_{cage}} \tag{9}$$

The elastic barrier is calculated by summing the harmonic elastic energy of all particles outside the cage, the center of which lies a distance z from the interface. The elastic barrier in the lower space is thus:

$$F_{el}^{(1)} = 2\pi\rho \frac{K_0 \Delta r_{eff}^2}{2} r_{cage}^4 \left[\int_0^{r_{cage}} dh \int_{\sqrt{r_{cage}^2 - h^2}}^{\infty} \frac{sds}{\left(s^2+h^2\right)^2} + \int_{r_{cage}}^{z} dh \int_0^{\infty} \frac{sds}{\left(s^2+h^2\right)^2}\right] = \frac{F_{el}^{bulk}}{2}\left[1 - \frac{r_{cage}}{2z}\right] \tag{10}$$

where K_0 is the spring constant of the bulk fluid localized state (minimum of the dynamic free energy) and F_{el}^{bulk} is the bulk fluid elastic barrier. To obtain the final analytical expression in equation (10) appropriate for distances far from the interface we employed $K_0(z) \approx K_{0,bulk} \equiv K_0$ and $\Delta r_{eff}(z) \approx \Delta r_{eff,bulk} \equiv \Delta r_{eff}$. The elastic barrier reduction relative to the bulk far from the interface originates only from the vapor interface-induced sharp truncation of the collective elastic displacement field. Similarly, the elastic barrier in the upper space is

$$F_{el}^{(2)} = 2\pi\rho \frac{K_0 \Delta r_{eff}^2}{2} r_{cage}^4 \left[\int_0^{r_{cage}} dh \int_{\sqrt{r_{cage}^2 - h^2}}^{\infty} \frac{sds}{\left(s^2+h^2\right)^2} + \int_{r_{cage}}^{\infty} dh \int_0^{\infty} \frac{sds}{\left(s^2+h^2\right)^2}\right] = \frac{F_{el}^{bulk}}{2} \tag{11}$$

Thus, the net collective elastic barrier for z>>d is:

$$F_{el}(z) \equiv F_{el} = F_{el}^{(1)} + F_{el}^{(2)} = \frac{F_{el}^{bulk}}{2}\left[1 - \frac{r_{cage}}{2z}\right] + \frac{F_{el}^{bulk}}{2} = F_{el}^{bulk}\left[1 - \frac{r_{cage}}{4z}\right] \tag{12}$$

In general (for all z), the mean alpha relaxation time is identified with the mean Kramer's time to cross the barrier of the dynamic free energy (which now depends on distance from the interface) and is given by[18,19]:

$$\frac{\tau_\alpha(z)}{\tau_s} = 1 + \frac{2\pi}{\sqrt{K_0(z) K_B(z)}} \exp\left(\frac{F_B(z) + F_{el}(z)}{k_B T}\right) \tag{13}$$

where τ_s is a known short time scale[15,16] and $K_B(z)$ is the absolute value of the curvature at the barrier position of $F_{dyn}(r,z)$. When probing relaxation deep in the film interior in the supercooled state with a glass transition temperature defined by the vitrification time scale τ_c, one has $K_B(z) \approx K_{B,bulk} \equiv K_B$, and thus

$$\tau_c \approx \frac{2\pi\tau_s}{\sqrt{K_0 K_B}} \exp\left(\frac{F_B(z) + F_{el}(z)}{k_B T_g(z)} \right) \tag{14}$$

or

$$T_g(z) \approx \frac{F_B(z) + F_{el}(z)}{k_B \ln\left(\frac{\tau_c \sqrt{K_0 K_B}}{2\pi\tau_s} \right)} \tag{15}$$

Since $F_{dyn}(z)$ exponentially decays with increasing z to its bulk form[18], $F_B(z)$ is expected to have an exponential gradient form, as previously shown numerically and rationalized analytically[16]. On the other hand, since the elastic displacement field outside a cage decays spatially as an inverse power law, $F_{el}(z)$ has an inverse power law dependence on z far from the interface. At large z, since $F_B(z) \rightarrow F_B^{bulk}$, the long range power law tail of $F_{el}(z)$ is fully responsible for spatial gradient dependence of $T_g(z)$, and one obtains:

$$\frac{T_{g,bulk} - T_g(z)}{T_{g,bulk}} \approx \frac{F_{el}^{bulk} - F_{el}(z)}{F_B^{bulk} + F_{el}^{bulk}} = \frac{F_{el}^{bulk}}{F_B^{bulk} + F_{el}^{bulk}} \frac{r_{cage}}{4z} \tag{16}$$

Equation (16) corresponds to a power law decay of the gradient of the local glass transition temperature far from the vapor-liquid interface, and is in excellent accord (as it should be) with the numerical calculations reported in the main text.

Given the glass transition temperature is determined from the alpha relaxation time, and given deviations from bulk behavior are perturbative at the large distances from the interface relevant to the long range gradient tail, the *same* analysis as above immediately implies the logarithm of the alpha time gradient far from the interface also has the same z^{-1} tail. This deduction has been explicitly verified in our numerical calculations reported in the main text.

Numerical Details of Theory Calculations and Mapping of Thermal Liquids to Effective Hard Sphere Fluids

Concerning quantitative numerical aspects of the full ECNLE theory results reported in Figures 1 and 3 of the main text, we first note that ECNLE theory is applicable to any fluid of spherical particles. In practice, in its numerical implementation for the hard sphere fluid reported in this article we employed a very small step size in Φ equal to 0.0001, which generates 10 times more data points of $\tau_\alpha(z,\Phi)$ compared to previous work [10,16]. Then, the volume fraction of the bulk glass transition, $\Phi_{g,bulk}$, and its film analog gradient, $\Phi_g(z)$, are found by taking $\tau_\alpha(z \rightarrow \infty, \Phi_{g,bulk})$ and $\tau_\alpha(z, \Phi_g)$, respectively, to be equal to a specified vitrification time (1 ns, 100 ns, 100 s are shown in the main text), and the predictions of the theory are determined using an iterative numerical method.

To perform quantitative calculations for real thermal liquids, a mapping between these systems and a reference hard sphere fluid is employed based on requiring the thermodynamic dimensionless compressibility (proportional to the dimensionless amplitude of long wavelength thermal density fluctuations) of any specific real liquid (determined from experimental equation-of-state data) exactly equals that of the reference hard sphere fluid. This results in a system-specific quantitative link between temperature and effective volume fraction given by[15,17,18]

$$T(\Phi) = \frac{B}{A + \frac{1}{\sqrt{N_S \frac{(1-\Phi)^4}{(1+2\Phi)^2}}}}, \tag{17}$$

where N_s is the known number of interaction sites of a molecule or Kuhn segment of a polymer, and A and B are parameters determined from the experimental equation-of-state of the real thermal liquid. Cooling under constant pressure conditions increases the density of the effective hard sphere fluid. For polystyrene, a typical fragile glass forming liquid, at 1 atm pressure it is known that $N_s = 38.4$, $A = 0.618$ and $B = 1297\ K$. [15,17,18]

Since the calculated values of $\Phi_{g,bulk}$ and $\Phi_g(z)$ have 5 digits of accuracy, using the thermal mapping (equation (17)) allows us to obtain the same levels of accuracy in computing $T_{g,bulk}$ and $T_g(z)$. Thus, the numerical accuracy of the theoretical predictions for the T_g gradient presented in the main article have been improved significantly relative to prior studies[18] in order to determine the long range tail of this quantity much more accurately. We have found that lower levels of numerical accuracy in the calculation can lead to inaccurate results for the form of the long range power law tail. Our improved numerics in this work yield predictions for the tail in full agreement with the asymptotic analytic results derived above.

Nonuniversality of the Amplitude of the Long Range Tail of Dynamic Gradients

Finally, we emphasize that equation (16) predicts a *universal* form for the power law tail as a consequence of generic long range collective elastic effects. However, based on extensive studies of real world glass forming molecular and polymer liquids using ECNLE theory[15,17,19,22], the amplitude of the tail is *nonuniversal* via the system-specific value of the ratio of the elastic to bulk local cage barriers in the *bulk* liquid, $R \equiv F_{elastic}/F_B$.

To address nonuniversal effects, we first note that, as is well established, for a fixed chemical system or simulation model, if the vitrification time scale is increased (e.g., simulation vs typical experiments), then the collective elastic effects predicted by ECNLE theory in the bulk become more important in a relative sense, i.e., the ratio *R* increases[15–17,22] This implies the amplitude of the long range tail will grow. Second, quantitative applications of bulk ECNLE theory to a chemically diverse set of polymer melts has established that[15] the nonuniversal dynamic fragility, m, grows linearly with R. Hence, the theory predicts that more fragile systems (larger values of m) have larger collective elastic effects, and hence a larger amplitude of the long range tail for the interface problem of present interest. Finally, older ECNLE theory work for bulk liquids based on simple models[15–17], and recent ECNLE-theory inspired analysis of the alpha relaxation behavior of molecular, polymer, and inorganic network glass forming liquids[22], have argued (and provided experimental evidence for) that below a fragility of ~ 30-40 the collective elastic effects become negligible in the bulk. Hence, in this “strong” glass former limit, we expect the amplitude of the long range tail will become extremely small in a practical sense or vanish. Future experimental and simulation studies can test the predicted quantitative nonuniversality of the amplitude of the long range tail, and the universality of its predicted 1/z form.

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